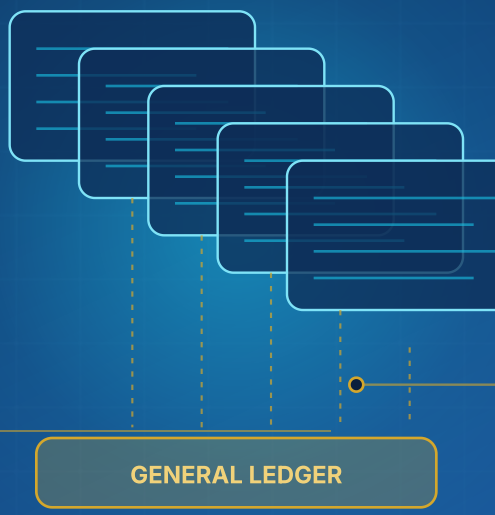
RBC CAPITAL MARKETS · DIGITAL ASSET TECHNOLOGY

Digital Asset Subledgers

The value to Finance of Line of Business subledgers that connect 24/7 chains to the General Ledger: reconciliation, intraday visibility, a faster close, cleaner controls and lower operating cost.

- 0 Summary first
- 1 The subledger concept
- 2 Connecting the Lines of Business; dedicated vs shared
- 3 The value to Finance
- 4 Operating model
- 5 Risks and controls
- 6 Next steps and open questions
- A Appendices: finality, journals, data model



Head of Digital Asset Technology, RBC Capital Markets

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Position paper for discussion. Not accounting, legal or regulatory advice. Items tagged VERIFY need confirmation with Finance Policy, Legal, Regulatory Reporting or the network operator before they are relied on. No financial benefit figures are asserted; bracketed gold placeholders mark numbers Finance should baseline. Diagrams and journal examples are illustrative design patterns, not descriptions of systems in production.

SUMMARY FIRST

0 Digital asset subledgers at a glance

Chains run 24/7 and settle in seconds or minutes. The General Ledger (GL) is an aggregate, legal-entity book built around periods, controls and a batch close. **Real time belongs in the Line of Business subledgers and the Capital Markets intraday core, not in the GL.** Each subledger is an event-sourced, reconciled, real-time operational book that posts summarized, rule-driven journals to the GL on defined windows.

7
Lines of Business, each with a dedicated or shared subledger, its own policy and reconciliation

1
common central path to the two Core services: Fort Knox and Tokenized Deposits

3-way
reconciliation: chain or ledger ⇄ subledger ⇄ GL, continuous and at every posting window

24/7 → close
per-entity cut-off turns continuous activity into closed accounting days

Theme	Position	What Finance gets
Book of record	The GL stays the aggregate book of record. LoB subledgers are the operational and position books of record. A chain is not a book of record, but a token ledger can be the golden source for holdings.	A GL that is not asked to behave like a chain; drill-down that substantiates every GL balance.
Connecting the Lines of Business	Each Line of Business owns its pipeline, node connectivity, posting rules and reconciliation. Shared standards, not shared execution: no central Kafka hub, shared execution hub or central hot KMS.	One chart-of-accounts mapping and one event schema across seven Lines of Business, so consolidation works without re-mapping.
Dedicated or shared subledger	A Line of Business can run its own subledger or share one, partitioned by legal entity, Line of Business, book and product, with segregated CoA segments. Execution, signing, policy, approvals and GL mapping stay with each business.	Fewer reconciliations and reuse where products, legal entity or cash legs overlap, without a shared execution hub.
Core services	Fort Knox cold custody (Investor Services / EISG) and the Tokenized Deposits ledger are reached through one common central path. Their events are authoritative and need no mirror wait.	No bespoke custody or Tokenized Deposits integrations per Line of Business; one completeness proof for every consumer.
Finality	Deterministic networks (internal permissioned ledger, Canton Network, Project Agora as designed VERIFY) post on commit. Probabilistic chains (Ethereum, Solana) hold a shadow balance until final.	The subledger handles network finality internally; Finance and the GL only ever receive final, authoritative entries.
Reconciliation	Three-way, continuous at the subledger, and again at each GL posting window, with a break taxonomy and maker-checker resolution.	Breaks found close to when they occur, not at month-end; fewer manual reconciliations and suspense items.
Intraday visibility	Shadow and authoritative balances per position; intraday provisional journals where intraday positions matter.	Intraday view of tokenized-deposit liabilities, settlement and financing positions for Treasury and liquidity reporting.
Close	Accounting date derived from finality time against a per-entity cut-off; closed, immutable event sets per day.	A close that does not wait on chain-by-chain investigation; weekend activity bucketed and posted on the next business day.
Controls and audit	Append-only, tamper-evident event log; every journal line traceable to events, rule version and Watchtower decision ID.	ICFR evidence (NI 52-109, SOX 404 where applicable) produced by design rather than assembled after the fact.
Capital and liquidity	Basel / OSFI classification attributes stored per position and versioned. OSFI's crypto-asset guideline (2027) takes effect 1 November 2026 for October year-end banks VERIFY.	A rule change becomes a reference-data change, not a re-platforming of seven Lines of Business.
Cost	Standards reused across Lines of Business; deterministic, replayable journals; automated break classification.	Lower manual effort in reconciliation, journal adjustment, intercompany matching and audit support [baseline TBD]; metrics in Section 3.8.

SECTION 1 · THE CONCEPT

1 Why a subledger sits between chain and GL

Three different records are in play, and most confusion in digital-asset accounting comes from treating them as one. The **chain** records token state. The **subledger** records what that state means for the bank. The **GL** records the aggregate financial position by legal entity and period.

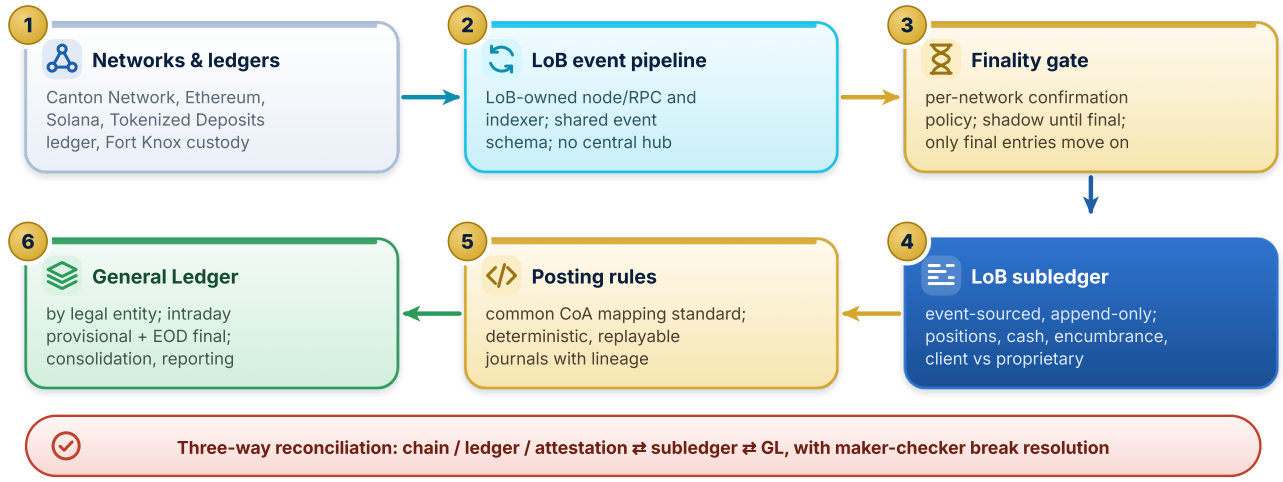


Figure 1. The subledger journey. Events move from networks to the GL through a Line of Business pipeline, a finality gate, an event-sourced subledger and deterministic posting rules. Reconciliation spans all three records.

The General Ledger

Aggregate book of record by legal entity, account and period. Supports financial statements, regulatory capital and liquidity returns, and ICFR attestations (NI 52-109 in Canada; SOX section 404 for SEC-registered reporting). It is not designed for wallet movements, block-level events or client-level token holdings, and should not be.

The Line of Business subledger

Holds the detail the GL leaves out: positions by instrument, holder and location; cash and tokenized-deposit balances; collateral and encumbrance; pending versus settled; fees and gas; and the client-versus-proprietary split. Produces GL journals and is the drill-down layer auditors and regulators use.

The chain or token ledger

Records token state and transfers. It does not record accounting classification, legal-entity attribution, beneficial ownership in omnibus structures, accruals, valuation or impairment, and cannot tell bank assets from client assets. It can be the **golden source for holdings**, but not the book of record.

Golden source versus book of record

- **Internal ledgers are authoritative.** The Tokenized Deposits ledger and the internal permissioned ledger partitions (cash / tokenized deposits, asset / RWA, encumbrance) are operated by the bank. The Global Transaction Banking subledger is a faithful projection of the Tokenized Deposits ledger. Tokenization does not change the legal character of the deposit: it remains a deposit liability; only the recording technology changes (consistent with Project Agora's findings).
- **External networks are mirrored and attested.** Canton Network, Ethereum, Solana, stablecoin rails and digital securities depositories are outside the bank's control. The subledger mirrors on-chain state for bank-controlled addresses or parties, confirmed by an independent attestation: signed custody statements, participant-node confirmations, depository statements or independently queried chain snapshots.
- **Legal register can differ from the token ledger.** For digital securities issued through a depository or registrar, the legally recognized register may not be the raw token ledger. The legal golden source is set per instrument. VERIFY

One subledger per business, or a shared one

The subledger is a logical book, not necessarily a separate system. A Line of Business can run a **dedicated** subledger, or several Lines of Business can **share** one subledger platform, each in its own partition (legal entity, Line of Business, book, product code) with segregated chart-of-accounts segments and its own maker-checker. Either way the same standards apply, and execution and signing stay with each business. Section 2.2 sets out when sharing makes sense.

Four design properties that create the value

Event-sourced DESIGN

Append-only log of normalized business events; balances are projections. Replay rebuilds state after a defect or a rule change.

Idempotent DESIGN

Deterministic key from native coordinates (network + transaction or update ID + event index, plus legal entity and book). Retries cannot double-post.

Bitemporal DESIGN

Each event carries event time (chain), knowledge time (when the bank saw it) and accounting date (after cut-off).

Finality-aware DESIGN

Shadow balance until a network-specific finality condition is met; only authoritative events generate journals.

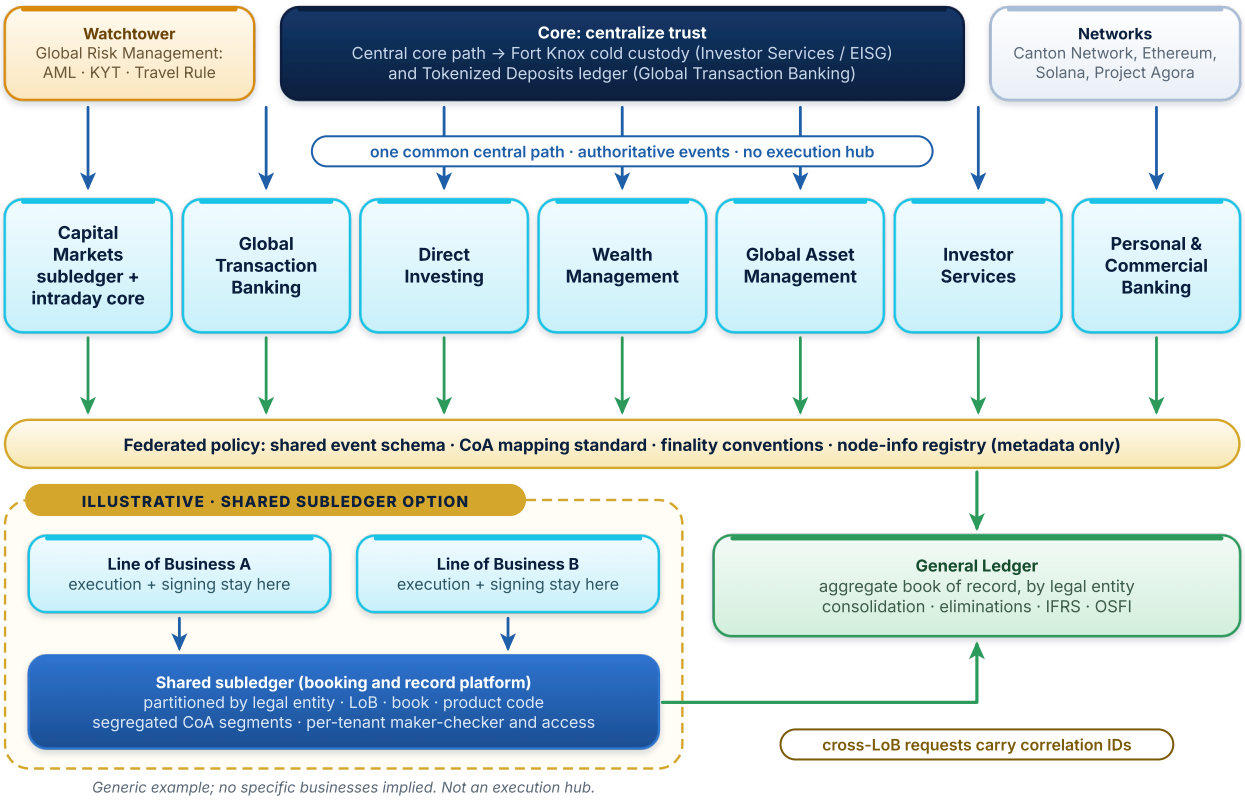
WHAT IT MEANS FOR FINANCE

Finance does not need to learn every chain. The subledger handles network finality internally; Finance and the GL only ever receive final, authoritative entries. It receives journals from a layer that has already applied finality, classification and legal-entity attribution, and that can show, line by line, which on-chain events produced each GL number.

SECTION 2 · CONNECTING THE LINES OF BUSINESS

2 One standard, dedicated or shared books, no execution hub

The subledger layer follows Core-Edge: **centralize trust, federate policy, decentralize execution**. Trust services sit in the Core and are reached through one path. Policy is federated as shared standards. Each Line of Business executes and keeps its own book, in a dedicated subledger or its own partition of a shared one. There is no shared execution hub and no central client data record.



Generic example; no specific businesses implied. Not an execution hub.

Figure 2. How the subledgers connect. Every Line of Business reaches Fort Knox and the Tokenized Deposits ledger through the same central path, conforms to the same standards and posts to the GL by legal entity. The inset is an illustrative example of two Lines of Business sharing one partitioned subledger; it does not indicate which businesses share. Cross-LoB requests carry correlation IDs, not client identity.

What is centralized (trust)

- One common central path to Fort Knox cold custody and the Tokenized Deposits ledger, with a single event and attestation format.
- One place where the Core services prove completeness (sequence numbers or offsets) to every consuming Line of Business.
- Network confirmation thresholds approved centrally and held in the node-info registry (metadata only).

What is federated (policy) and decentralized (execution)

- Shared: event envelope and schema catalogue, CoA mapping standard, idempotency and finality conventions, ISO 20022 mapping for payment-type events.
- Owned by each Line of Business: event pipeline, node / RPC connectivity, posting rules and GL mapping, reconciliation instance (or its partition of a shared one), hot or warm signing for external networks.
- Failure domains stay within a Line of Business.

Line of Business	What the subledger tracks	Golden source(s)	GL posting cadence
Capital Markets	Trading and inventory positions in tokenized securities and approved crypto assets; settlement state; repo and collateral encumbrance; tokenized-deposit and stablecoin cash legs; financing; fees; network gas	Intraday core; external chains via mirror + attestation; Tokenized Deposits ledger for cash legs	Intraday provisional at defined windows; EOD final with revaluation
Global Transaction Banking (Global Transaction/Cash Management)	Tokenized deposit liabilities by client; mint / burn; locks and holds; Project Agora settlement positions and tokenized nostro balances; fees; interest accrual if tokenized deposits bear interest	Tokenized Deposits ledger (authoritative); core banking DDA; Project Agora platform statements	Near-real-time or intraday (liquidity relevance); EOD final
Direct Investing	Client holdings in Fort Knox wallets; client cash and encumbrance for pending orders; commissions; pending fulfillment with Capital Markets	Fort Knox custody records via the central path; Direct Investing books for client entitlement	EOD for revenue and client cash; memo / off-balance-sheet records daily
Wealth Management	Client positions in tokenized funds and securities; advisory and platform fees; suitability-linked holdings	Fund register / transfer agent or custodian (often Investor Services); tokenized fund ledger where it is the legal register VERIFY	EOD (fees, client cash); off-balance-sheet records daily
Global Asset Management	Bank side: management fees, seed capital, co-investment in tokenized funds. Fund side (separate fund books): unit register, NAV, subscriptions and redemptions	Fund administrator NAV; on-chain unit register where it is the legal register VERIFY	Bank GL: daily / EOD fee accruals and seed positions. Fund books: NAV cycle
Investor Services (custody, Fort Knox operator)	Assets under custody by client and location; asset servicing (coupons, corporate actions); custody fees; attestations issued	Custody books of record, reconciled to on-chain wallet balances	Daily fee accruals; off-balance-sheet AUC daily; real-time custody state
Personal & Commercial Banking	Client use of tokenized deposits where offered VERIFY; link to conventional deposits; fees	Core banking; Tokenized Deposits ledger	EOD (standard banking close)

2.1 Where the Lines of Business meet

Cross-LoB flows are where value and risk concentrate, because they usually cross legal entities. Each side posts its own entries with a shared intercompany reference carried in the event (fulfillment ID, settlement ID). Because there is no central client data record, that reference is a transaction-level correlation key, not a client identifier.

Direct Investing → Capital Markets FULFILLMENT

Direct Investing keeps the client relationship and entitlement; Capital Markets records its principal or agency leg plus an intercompany receivable or payable where legal entities differ.

Capital Markets → Global Transaction Banking CASH LEG

Capital Markets calls Global Transaction Banking for a tokenized-deposit cash leg or a Project Agora payment; Global Transaction Banking fulfils and books the liability side.

Investor Services → all CUSTODY

Investor Services keeps the custody book of record and issues the attestations other Lines of Business rely on; custody fees are charged across Lines of Business VERIFY.

Watchtower → all GATE

Every movement carries a Global Risk Management Watchtower decision ID; subledgers reject authoritative posting of outbound movements without one.

WHAT IT MEANS FOR FINANCE

- Intercompany matching runs on a key both sides already carry, so consolidation eliminations (IFRS 10) are mechanical rather than investigative.
- Gas and network fees are tracked by network, wallet and transaction, so they can be expensed or recharged to the requesting Line of Business.

2.2 Dedicated vs shared subledger

Some Lines of Business may share a subledger. A shared subledger is a **shared booking and record platform**, not a shared execution hub: each business still executes, signs and approves on its own, and keeps its own policy and GL mapping. Sharing changes where records are kept, not who controls them. Whether any two businesses should share is a decision for those businesses and Finance; this paper does not assume any particular pairing.

When sharing makes sense

- **Similar products**: the same instrument classes, event types and posting patterns.
- **Same legal entity or booking model**: activity books to the same entity, or follows the same principal / agency model.
- **A shared cash leg**: for example Global Transaction Banking (Global Transaction/Cash Management) providing the tokenized-deposit cash leg for other businesses, where both sides would otherwise reconcile the same movements.

How it is partitioned

- Every record keyed by **legal entity, Line of Business, book and product code**.
- **Segregated chart-of-accounts segments** per business, so GL keys never mix.
- **Per-tenant maker-checker and access controls**: no user or service of one business can view, post or approve in another's partition.
- Per-tenant reconciliation views and break queues.

What stays federated

- Each business keeps its own **policy**, **approvals** and **GL mapping** (posting-rule versions per tenant).
- Execution, signing keys and node connectivity stay with each business, consistent with Core-Edge.
- Client identity stays in each business's own domain; the shared platform holds Line of Business-scoped references only.

What Finance gets

- **Fewer reconciliations**: a movement both businesses touch (for example a shared cash leg) is recorded once, with two partitioned views, rather than reconciled between two books.
- **Reuse**: one platform build, one set of conformance tests and one operations team.
- **Lower run cost** than separate platforms for small or similar businesses [baseline TBD].

Blast radius

An outage, defect or bad release affects every tenant at once.

MITIGATION

Per-tenant replay and isolation; staged releases; defined degraded mode per tenant.

Change coordination

One business's change waits on another's testing and release window.

MITIGATION

Per-tenant posting-rule and mapping versions; shared release calendar; conformance tests.

Entitlement leakage

Data or posting rights cross partitions, including client information.

MITIGATION

Tenant-scoped access and maker-checker; access reviews; reconciliation and key events by partition.

Criterion	Favours dedicated	Favours shared
Products and event types	Distinct instruments or lifecycles	Same instrument classes and posting patterns

Legal entity / booking model	Different entities or booking models	Same entity or same principal / agency model
Cash leg	No common cash leg	Shared cash leg (e.g. Global Transaction/Cash Management for other businesses)
Volume and change pace	High volume or frequent, independent change	Modest volume; changes can be co-ordinated
Regulatory or client-data separation	Hard separation required (e.g. fund books vs bank books) VERIFY	Partitioning by entity, book and access control is sufficient VERIFY
Resilience appetite	Outage of one must not affect the other	A shared outage is acceptable with per-tenant degraded mode
Platform ownership	No natural host	A clear host and platform owner can be named

Criteria are qualitative and for discussion. Any sharing decision would be confirmed by the businesses involved, Finance and Risk.

SECTION 3 · THE VALUE TO FINANCE

3 Seven levers, from reconciliation to cost

The value of the subledger layer to Finance does not come from blockchain itself. It comes from moving reconciliation, classification and finality decisions **upstream**, to the point where events happen, so that what reaches the GL is already clean, explained and traceable. Each lever below compares a set-up without a subledger layer to one with it. Benefits are stated qualitatively; no figures are claimed.

3.1 Reconciliation

WITHOUT A SUBLEDGER LAYER

- Positions reconciled after the fact, often at month-end, against exports from several chains and custodians
- Breaks surface late and arrive unclassified
- Manual matching of chain data to GL accounts

WHAT THE SUBLEDGER DOES

- Three legs: chain / ledger / attestation vs subledger (continuous, per finalized block or offset); subledger vs GL (each posting window and EOD); closure check after explained timing differences
- Break taxonomy: timing, finality, mapping, completeness, valuation, fee / gas, unauthorized

WHAT FINANCE GETS

- Breaks found close to when they occur, with a type, owner and ageing clock
- Fewer suspense balances and manual adjustments
- Client-entitlement shortfall vs custody raised as a key event, not discovered at audit

3.2 Intraday position and cash visibility

WITHOUT A SUBLEDGER LAYER

- End-of-day view only; intraday tokenized activity invisible to Treasury
- Pending outbound transfers can be counted as available

WHAT THE SUBLEDGER DOES

- Every position carries a shadow balance (observed, plus pending outbound) and an authoritative balance (final)
- Conservative view: outbound pending reduces availability at once, inbound counts only when final
- Intraday provisional journals where intraday positions matter

WHAT FINANCE GETS

- Intraday view of tokenized-deposit liabilities for intraday liquidity
- Capital Markets settlement and financing positions visible as they change
- Limits and liquidity decisions on a conservative, explainable number

3.3 Close acceleration

WITHOUT A SUBLEDGER LAYER

- 24/7 activity forced into a batch calendar by hand
- Late-arriving or unconfirmed events reopen balances
- Weekend activity waits for Monday investigation

WHAT THE SUBLEDGER DOES

- Per-entity cut-off timestamp; accounting date derived from finality time
- Closed, immutable event set per accounting date while the subledger keeps processing
- Weekend accrual buckets posted on the next business day with original accounting dates VERIFY

WHAT FINANCE GETS

- A close that starts from reconciled, final data
- Provisional entries always reverse or true up, so close is never computed on unfinal data
- Fewer late adjustments [days-to-close baseline TBD]



Figure 3. From 24/7 to a closed accounting day. The subledger never stops; the GL receives a closed, immutable set of events for each accounting date.

3.4 GL posting

WITHOUT A SUBLEDGER LAYER

- Pressure to open a GL account per token or network
- Journal logic embedded in spreadsheets or one-off scripts

WHAT THE SUBLEDGER DOES

- CoA mapping standard maps subledger attributes to GL keys: legal entity; Line of Business and book; instrument class; accounting classification; prudential classification; position state; ownership; currency
- Network and contract address stay in the subledger as analytical dimensions
- Deterministic, replayable journals: same events + rule version + reference data = same journals

WHAT FINANCE GETS

- A GL chart that does not grow with every new token
- Journals that can be regenerated and proven
- Accounting treatment follows the instrument, not the technology (IFRS 9 for tokenized securities and deposits; IAS 2 / IAS 38 for crypto assets without contractual rights; ASU 2023-08 for US GAAP entities)

3.5 Controls and audit trail

WITHOUT A SUBLEDGER LAYER

- Evidence assembled after the fact from logs and screenshots
- Limited lineage from a GL balance back to on-chain activity

WHAT THE SUBLEDGER DOES

- Append-only, hash-chained event log
- Every journal line carries source event IDs, posting-rule version, reference-data version and Watchtower decision ID
- Maker-checker on manual journals, break resolutions, compensating transactions, mapping and rule changes

WHAT FINANCE GETS

- ICFR key controls documented by design (posting rules, finality gates, reconciliation, central-path feed)
- Audit drill-down from GL to subledger to chain in one path
- Segregation of duties between key custodians, operations, Finance, reconciliation owners and engineering

3.6 Capital and liquidity

WITHOUT A SUBLEDGER LAYER

- Prudential classification decided in reporting, far from the position
- A change in rules means re-mapping across systems

WHAT THE SUBLEDGER DOES

- Classification attributes held per position and versioned: network type, issuer and its status, stabilization mechanism, redemption rights, legal-finality assessment
- Basel group (1a, 1b, 2a, 2b) derived from reference data

WHAT FINANCE GETS

- Readiness for OSFI's Capital and Liquidity Treatment of Crypto-asset Exposures (Banking) Guideline (2027), effective 1 November 2026 for October year-end banks VERIFY
- Outcomes of the Basel Committee's targeted cryptoasset review absorbed as a rule-version change VERIFY
- Client vs proprietary split that supports on- / off-balance-sheet treatment and client-asset segregation

3.7 Cost and operational efficiencies

Build once, conform many — REUSE
One event schema, CoA mapping standard and reconciliation design reused by seven Lines of Business, instead of seven bespoke designs.

No bespoke Core integrations — REUSE
Lines of Business do not each build their own Fort Knox or Tokenized Deposits integrations; the central path provides one format and one completeness proof.

Replay instead of rework — AUTOMATION
A defect or rule change is fixed by replaying events into a new projection, not by manual journal correction.

Fewer manual journals — AUTOMATION
Deterministic posting rules remove spreadsheet journals; manual entries become exceptions under maker-checker.

Intercompany without disputes — AUTOMATION
Shared correlation keys make matching and elimination mechanical.

Cheaper audit support — EVIDENCE
Lineage from GL line to on-chain event reduces sampling effort and evidence requests.

Regulatory change as data — CHANGE
New Basel or OSFI rules are applied as versioned reference data rather than a re-platforming.

Contained failure — RESILIENCE
No central hub means an outage in one Line of Business pipeline does not stop the others.

3.8 What to measure before quoting a number

The paper deliberately makes no quantified benefit claim. The metrics below are what Finance would baseline today and track through a pilot. Every value cell is a placeholder.

Lever	Metric	Baseline	Target
Reconciliation	Manual reconciliation effort per month for digital-asset positions	[TBD]	[TBD]
Reconciliation	Open breaks older than one business day, by break type	[TBD]	[TBD]
Close	Business days from period end to digital-asset subledger sign-off	[TBD]	[TBD]
GL posting	Share of digital-asset journals that are manual or adjusted after posting	[TBD]	[TBD]
Intraday visibility	Lag between on-chain finality and Treasury visibility of the position	[TBD]	[TBD]
Controls	Audit evidence requests and time to fulfil for digital-asset balances	[TBD]	[TBD]
Intercompany	Unmatched intercompany items at close for cross-LoB flows	[TBD]	[TBD]
Cost	Run cost per Line of Business onboarded (build and operate)	[TBD]	[TBD]

SECTION 4 · OPERATING MODEL

4 Who owns what, and how a day runs

The operating model keeps execution at the Edge and Finance in control of accounting. Each business's book can be a dedicated subledger or a partition of a shared one (Section 2.2); the roles below are the same either way. The Capital Markets intraday core runs at trading speed and is not a GL; the subledger adds the accounting view; Finance owns posting rules, journal approval and the close.

Role	Owns	Does not own
Core (centralize trust)	Central core path; Fort Knox cold custody and its signing keys (Investor Services / EISG); Tokenized Deposits ledger; completeness guarantees to consumers	Line of Business execution; LoB-to-LoB messaging; a central hot KMS
Shared standards (federate policy)	Event envelope and schema catalogue; CoA mapping standard; finality semantics; node-info registry (metadata only); conformance tests per release	Running anyone's pipeline
Each Line of Business (decentralize execution)	Event pipeline; node / RPC (own or contracted under OSFI B-10); subledger or its partition of a shared one, with its own policy, approvals and GL mapping; reconciliation instance; break ownership; hot / warm signing for external networks	Client identity outside its own domain
Shared subledger platform owner (where used)	Platform operation, tenant isolation, per-tenant access and maker-checker enforcement, release calendar, per-tenant degraded mode	Any tenant's policy, approvals, GL mapping, execution or signing
Finance	Posting rules and their versions; journal approval; cut-off definitions; close; consolidation and eliminations; accounting policy	Chain connectivity
Global Risk Management Watchtower	Pre-movement gates (AML / KYT, sanctions, Travel Rule, policy-as-code limits); post-movement monitoring; key-event escalation	Accounting outcomes

Three journeys that show the model working

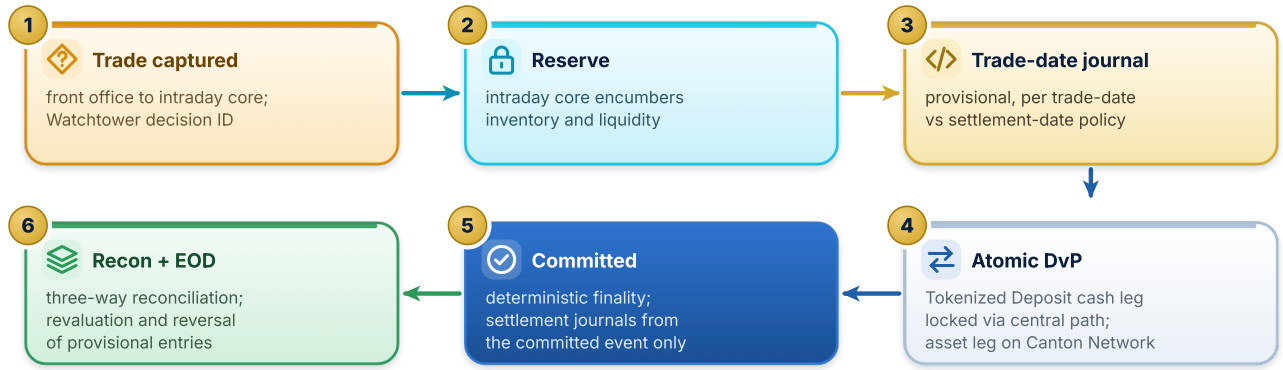


Figure 4. Capital Markets DvP. The trade-date journal is provisional and depends on RBC's trade-date vs settlement-date policy election VERIFY. Settlement journals come only from the committed settlement event.



Figure 5. Direct Investing to Capital Markets fulfillment. Client entitlement never runs ahead of custody: the Direct Investing subledger credits the holding only on Fort Knox confirmation.



Figure 6. Global Transaction Banking tokenized deposit lifecycle. Reclassification between conventional and tokenized deposit liabilities has no P&L effect, subject to policy confirmation VERIFY.

Running 24/7 against a batch GL

- **Two posting cadences.** Intraday provisional journals aggregated by accounting key at defined windows; EOD final journals at the entity's cut-off with revaluation, accruals and true-ups. Window frequency is a per-LoB design parameter.
- **Break management.** Each break has an owner, an ageing clock and escalation thresholds set by Line of Business policy. Unauthorized or unexplained movement always goes straight to Watchtower and Line of Business risk.
- **Valuation.** Tokenized deposits at par plus accrued interest; tokenized funds at administrator NAV; tokenized securities priced as their conventional equivalents; crypto assets and stablecoins from exchange or composite prices. On-chain oracles are one input among several, under independent price verification and IFRS 13.
- **Degraded mode.** If the central path is unavailable, Lines of Business hold events as pending and post nothing until completeness is proven on recovery VERIFY.

SECTION 5 · RISKS AND CONTROLS

5 What can go wrong, and the controls

Most of the risks below are not new to banking; they are reconciliation, completeness and classification risks that 24/7 networks make faster. Tile colours are an editorial guide to relative attention, not a formal risk rating. Further risks from the source paper (24/7 vs batch cut-off, LoB-owned hot keys, stablecoin issuer freeze or de-peg) are covered by the controls table.

Posting on non-final data

IMPACT
Positions and the GL would be misstated if an unconfirmed event were posted.

MITIGATION
The subledger handles network finality internally; Finance and the GL only ever receive final, authoritative entries. Finality conditions are approved centrally per network.

Client entitlement exceeds custody

IMPACT
Client-asset shortfall and regulatory breach.

MITIGATION
Continuous Leg-1 reconciliation per custody location; key event; segregation enforced at wallet level.

Missed or duplicated events

IMPACT
Node outage, offset gap or Canton participant disconnect understates movements; retries overstate them.

MITIGATION
Offset and height continuity checks; second independent node for attestation; deterministic idempotency keys.

Central-path outage

IMPACT
Lines of Business cannot confirm custody or tokenized-deposit state.

MITIGATION
Pending-not-posted rule; completeness proof on recovery; defined degraded mode VERIFY.

Classification drift

IMPACT
Basel review, OSFI guideline or a stablecoin status change misreports capital.

MITIGATION
Position-level, versioned classification attributes; reference-data-driven rules.

Inconsistent standards across Lines of Business

IMPACT
Reconciliation and consolidation failures.

MITIGATION
Shared schema catalogue, CoA mapping standard, conformance tests per release.

Oracle manipulation or stale prices

IMPACT
Valuation errors.

MITIGATION
Multi-source pricing, independent price verification, deviation and staleness thresholds, model governance (OSFI E-23 VERIFY).

Identity fragmentation

IMPACT
No central client record weakens cross-LoB monitoring.

MITIGATION
Correlation IDs on cross-LoB flows; Watchtower receives LoB-supplied attributes per policy; no shared identity store.

Control area	Design expectation
Maker-checker	Manual journals, break resolutions, compensating transactions on internal ledgers, encumbrance releases, CoA mapping and posting-rule changes, network confirmation parameters
Segregation of duties	Separate roles for key custodians (Fort Knox), Line of Business operations, Finance, reconciliation owners and engineering. No one person can both create a movement and approve its accounting
ICFR (NI 52-109 / SOX 404)	Posting rules, finality gates, reconciliation and the central-path feed treated as key controls with design and operating-effectiveness testing; IT general controls over each pipeline

Audit trail	Append-only log with hash-chaining or equivalent; every journal line traceable to events, rule and reference-data versions and Watchtower decision ID; retention per records policy VERIFY
Key events	Unauthorized movement; entitlement shortfall vs custody; stale or failed attestation; central-path offset gap; oracle deviation; stablecoin de-peg or issuer freeze; Watchtower override; GL suspense above threshold
Watchtower gates	Pre-movement AML / KYT, sanctions, Travel Rule data (FATF R.16; FINTRAC travel rule), policy-as-code limits; outbound movements without a valid decision ID are not posted
Technology and cyber	Pipelines, nodes and signing infrastructure in scope of OSFI B-13; third-party node / RPC providers under B-10; minimize hot balances and sweep to Fort Knox

SECTION 6 · NEXT STEPS

6 An indicative sequence and the decisions it needs

The sequence below is indicative and carries no dates or budget. It starts where the bank already controls the ledger (Tokenized Deposits) and where Finance value is most visible (Capital Markets settlement and intraday liquidity), then extends the same standards to the remaining Lines of Business.

PHASE 0

Decide and baseline

Answer the open questions on legal entities, policy elections and registers. Finance baselines the Section 3.8 metrics. Publish the event schema, CoA mapping standard and finality conventions.

PHASE 1

Prove on internal ledgers

Global Transaction Banking subledger projecting the Tokenized Deposits ledger via the central path; closing control ledger = subledger = GL; intraday provisional postings tested against GL capability.

PHASE 2

Add Capital Markets settlement

Capital Markets subledger downstream of the intraday core; DvP with a tokenized-deposit cash leg on a deterministic network; three-way reconciliation and break management live.

PHASE 3

Extend by conformance

Direct Investing and Investor Services (Fort Knox) next, then Wealth Management, Global Asset Management and Personal & Commercial Banking as product approvals allow, each on a dedicated subledger or a partition of a shared one per the Section 2.2 criteria. Probabilistic chains only once the subledger's finality handling is proven.

Decisions requested

- **Endorse the principle** that real time sits in Line of Business subledgers and the intraday core, and the GL receives summarized, rule-driven journals on defined windows.
- **Mandate the shared standards** (event schema, CoA mapping, finality conventions) as conditions for any Line of Business going live.
- **Name Finance owners** for posting rules, cut-off definitions and the Section 3.8 baseline.
- **Agree the Phase 1 scope** on the Tokenized Deposits ledger with Global Transaction Banking.

Open questions

1. Which legal entities book each Line of Business's digital-asset activity, and what intercompany agreements govern Direct Investing to Capital Markets fulfillment and Capital Markets to Global Transaction Banking cash legs?
2. What is RBC's trade-date vs settlement-date policy election for tokenized securities settling atomically?
3. For each tokenized instrument and fund, what is the legally recognized register, and is the on-chain ledger the legal golden source?
4. Do custodied client crypto assets in Fort Knox stay off balance sheet under RBC's IFRS policy, and how are safeguarding obligations assessed for any US GAAP entities after SAB 122?
5. Which stablecoins (if any) are approved rails, and how are they classified for accounting and for Basel / OSFI Group 1b vs Group 2?
6. How do section-level requirements of OSFI's 2027 crypto-asset guideline map to subledger attributes and reporting outputs, and what follows from the Basel review update?
7. Which confirmation policies (Ethereum 'finalized' vs 'safe'; Solana 'finalized' vs 'confirmed') are approved centrally per use case, and which Layer-2 networks, if any, are in scope?
8. Does the GL support provisional intraday postings and back-dated accounting dates within an open period for weekend activity?
9. What completeness guarantees (sequencing, replay window, degraded mode) does the central path give consuming Lines of Business?
10. What attributes may Lines of Business share with Watchtower under the pseudo-federated identity model?
11. Which Lines of Business, if any, should share a subledger, who hosts and owns a shared platform, and how are its run costs allocated?
12. Is staking or any network-reward activity contemplated? If so, what accounting policy applies?
13. What is RBC's role and timeline in Project Agora following the 2026 report and real-value testing?

APPENDICES

A Finality, illustrative journals and the common data model

A1. Finality handling by network

Confirmation thresholds are network parameters held in the node-info registry and approved centrally, not hard-coded per Line of Business. Layer-2 rollups should distinguish sequencer soft-confirmation from L1 settlement VERIFY.

Network	Finality model	Shadow → authoritative	Correction handling
Tokenized Deposits ledger; internal permissioned ledger partitions	Bank-operated; deterministic on commit VERIFY	Post on commit	Errors fixed by compensating transactions under maker-checker
Canton Network (privacy-enabled)	Deterministic: final once the synchronizer commits; sub-transaction privacy limits visibility to stakeholders	Post on commit observed at the LoB's participant node	Watch for participant-node lag or disconnect (offset gaps)
Ethereum	Probabilistic head; economic finality via Casper FFG checkpoints (typically about two epochs)	Shadow on inclusion; authoritative on 'finalized'	Handled inside the subledger; only final entries reach the GL
Solana	Commitment levels: processed, confirmed, finalized (rooted)	Shadow on confirmed; authoritative on finalized	Handled inside the subledger; only final entries reach the GL
Stablecoin rails	Underlying chain finality plus issuer controls (freeze / blacklist, off-chain mint / redeem)	Chain rule plus issuer confirmation for mint / redeem	Issuer freeze events monitored as key events
Project Agora (via Global Transaction Banking)	Atomic settlement with legal finality reported achievable VERIFY	On final settlement notice	Per platform rulebook VERIFY
Digital securities depositories	Legal finality per depository rulebook	On depository settlement confirmation	Per rulebook

A2. Illustrative accounting events

Entries are illustrative and need validation by RBC Finance Policy VERIFY. Account selection depends on the instrument's classification and accounting policy.

Event	Illustrative journal (Dr / Cr)	Notes
Tokenized deposit mint (issuer)	Dr Client deposit (DDA) / Cr Tokenized deposit liability	Reclass within deposit liabilities; no P&L VERIFY
Tokenized deposit burn / redeem	Dr Tokenized deposit liability / Cr Client deposit (DDA)	Reverse of mint
Tokenized deposit transfer	Dr TD liability (payer) / Cr TD liability (payee)	Inter-entity or interbank flows add nostro / settlement entries
Tokenized security buy (Capital Markets, FVTPL)	Trade date: Dr Trading asset / Cr Settlement payable. Settle: Dr Settlement payable / Cr Cash (TD / nostro)	Atomic same-day DvP may collapse the two dates
Atomic DvP / PvP	Asset and cash legs posted together under one settlement ID	PFMI Principle 12
Repo pledge / release	Dr Cash / Cr Repo liability; asset to 'pledged' sub-account. Release reverses, with interest expense	Pledged asset usually not derecognized
Coupon / distribution	Accrual: Dr Interest receivable / Cr Interest income. Receipt: Dr Cash / Cr Receivable	Custody holdings: client cash credit only

Gas / network fees	Dr Network fee expense (or recharge) / Cr Crypto asset held for fees	Measurement per IAS 38 / IAS 2 or ASU 2023-08
Staking (only if approved)	No assumed treatment	Policy needed VERIFY

A3. Minimum common data model

Each Line of Business implements these entities in its own subledger; names, keys and semantics are shared standards.

Entity	Key attributes
LegalEntity / LoB / Book	entity code, Line of Business, book, accounting cut-off, reference time zone
Network	network ID, type, finality model, confirmation policy version, approval status (from node-info registry)
InstrumentDefinition	instrument ID, network, contract / template ID, instrument class, accounting class, Basel / OSFI group, issuer, legal golden source (versioned)
Wallet / LedgerAccount	address or party ID, custody location, owner type (proprietary / client), segregation model
PartyRef	Line of Business-scoped party ID; no cross-LoB identity join
RawChainEvent → BusinessEvent	native coordinates and finality status → normalized event with idempotency key, correlation IDs, Watchtower decision ID, accounting date
Position / Encumbrance	state (shadow, authoritative, encumbered, in-transit); encumbrance type, counterparty, start / release
PriceObservation / Attestation	source, value, IFRS 13 level, IPV status; attestation scope, signature, independence flag
JournalEntry / JournalLine	legal entity, GL key, Dr / Cr, amount, currency, provisional / final, rule version, source event IDs
ReconRun / Break	leg, scope, result; break type, severity, owner, ageing, maker and checker

Prepared by the Head of Digital Asset Technology, RBC Capital Markets. Edited and expanded from the source paper "Digital Asset Subledgers by Line of Business: General Ledger integration and real-time synchronization with internal and external chains". Standards and guidance named in this paper (IFRS 9, IFRS 10, IFRS 13, IAS 2, IAS 38, IFRS Interpretations Committee 2019 agenda decision, ASU 2023-08, SAB 122, Basel SCO60, OSFI B-10, B-13 and E-23, OSFI Capital and Liquidity Treatment of Crypto-asset Exposures (Banking) Guideline, CPMI-IOSCO PFMI, FATF R.16, NI 52-109, SOX 404) are cited as described in the source paper; its numbered reference list was not included in the version provided and should be re-attached before circulation. Not accounting, legal or regulatory advice.